

q-ridge: certifying the empirical quadrature of a Burgers correction
ladder
on the states the reduced model actually reaches

2026-09-16

1 q-ridge – the regression is the quadrature rule; certifying it on reachable states removes almost all of it, but not the top rung within the constructible $m \leq 2048$

Final numbers. Three cluster jobs on the frozen Burgers checkpoint 18f0266ae6f0... at 256 intervals, on the six opened development cases, under the budget-600 block-damped variable-projection contract inherited from b-ladder-top. The pre-registered design and its amendment §A3, which re-scoped the lane after the q-diag lane reported, are in ./DESIGN.md.

job	attempt	question	job id	GPU	source commit	elapsed (s)	failed gates
1	qrg101	R1	3757235	NVIDIA A100 80GB PCIe	5fc6099127d4	5389.4	none
2	qrg201	R2	3757237	NVIDIA A100 80GB PCIe	5fc6099127d4	6834.0	none
3	qrg304	EQCERT	3768168	NVIDIA A100 80GB PCIe	467a4670f179	10502.3	none

1.1 Where the regression is, measured before anything was submitted

Recomputed from the retained cclad01 fields (job 3734098, archive SHA256 4c6fd589688a253f...), worst over the six cases, against that job’s own converged full-order solve. The q-diag lane reached the same conclusion independently and across four jobs.

configuration	$q = 0$	$q = 16$	$q = 32$	$q = 64$	$q = 128$	monotone in q ?
m4 dense	1.8890	1.3984	1.2336	1.0843	0.8930	yes
M=256 dense	1.2710	1.2305	1.1869	1.1255	1.0418	yes
M=256 EQ	1.3113	1.4956	1.5231	1.5557	1.6141	no
m4 EQ	1.9002	2.9937	–	–	–	no

The dense and the EQ arms at $M = 256$ solve the ****identical**** M test equations against the ****identical**** $K + q$ unknowns; the only difference is that the advection integral is replaced by an m -point rule. The dense ladder is monotone and the EQ ladder is not.

1.2 EQ rule certification – the primary

****Verdict: no.**** The incumbent rule’s ladder is non-monotone on the evolved metric; every rung of the rebuilt ladder certified at the primary bar: no; the top rung certified: no. No m in the declared

grid reaches the bar at the top rung; a log-log extrapolation of $\rho_{\max}$ against m puts the crossing at $m \approx 2522$ (slope -1.772).

1.2.1 The reachable-state population

q	M	iterates per step	fit states available	certification states available	collection (s)
0	64	13	15600	5200	23.8
16	128	13	15600	5200	27.1
32	192	13	15600	5200	32.8
64	320	13	15600	5200	43.6
128	576	13	15600	5200	66.5
256	1088	13	15600	5200	132.7

1.2.2 Certification: every rule, its NNLS fit and its held-out ρ

The bar is $\rho^* = 0.116$, declared in DESIGN.md §A3.4 before the job ran, from q-diag's measurement of the $q = 0$ incumbent rule. **A rule is never certified by its NNLS relative fit**, which is reported here precisely so the anti-correlation can be seen again on new rules.

q	population	m target	m achieved	m/M	NNLS relative fit	$\rho_{\max}$	ρ_{95}	ρ_{med}	certified	truncated
0	reachable	1024	1024	16.00	4.58e-05	0.0153	0.0088	0.0005	yes	no
0	reachable	2048	1521	23.77	7.83e-06	0.0261	0.0083	0.0002	yes	no
0	static	256	256	4.00	6.85e-03	0.2407	0.2073	0.0153	no	no
16	reachable	1024	1024	8.00	1.90e-04	0.0935	0.0273	0.0007	yes	no
16	reachable	2048	2048	16.00	1.92e-05	0.0452	0.0041	0.0002	yes	no
16	static	512	512	4.00	1.54e-03	0.1279	0.0523	0.0024	no	no
32	reachable	1024	1024	5.33	2.89e-04	0.0533	0.0226	0.0014	yes	no
32	reachable	2048	2048	10.67	2.66e-05	0.0668	0.0120	0.0002	yes	no
32	static	768	768	4.00	6.27e-04	0.1482	0.0214	0.0015	no	no
64	reachable	1024	1024	3.20	7.12e-04	0.0531	0.0230	0.0027	yes	no
64	reachable	2048	2048	6.40	7.54e-05	0.1120	0.0172	0.0010	yes	no
64	static	1280	1280	4.00	2.43e-04	0.1439	0.0235	0.0015	no	no
128	reachable	1024	1024	1.78	1.62e-03	0.2786	0.2246	0.0177	no	no
128	reachable	2048	2048	3.56	1.04e-04	0.1908	0.0518	0.0019	no	no
128	static	2048	2048	3.56	7.16e-05	0.2194	0.0677	0.0016	no	no
256	reachable	1024	1024	0.94	4.01e-02	0.5731	0.5370	0.1855	no	no
256	reachable	2048	2048	1.88	3.08e-04	0.1678	0.0957	0.0099	no	no
256	static	2048	2048	1.88	1.49e-04	0.4621	0.4275	0.0585	no	no

1.2.3 The rule chosen at each rung

q	M	chosen m	basis	$\rho_{\max}$	ρ_{95}	NNLS relative fit
0	64	1024	primary	0.0153	0.0088	4.58e-05
16	128	1024	primary	0.0935	0.0273	1.90e-04
32	192	1024	primary	0.0533	0.0226	2.89e-04
64	320	1024	primary	0.0531	0.0230	7.12e-04
128	576	2048	secondary	0.1908	0.0518	1.04e-04
256	1088	2048	secondary	0.1678	0.0957	3.08e-04

1.2.4 The dense ladder this is measured against

Read-only from the q-trajdirs lane, job 3757505 (qtd02) on NVIDIA A100-PCIE-40GB: the same six cases, the same old direction rule, the same budget-600 contract, exact quadrature, $M = 4(K + q)$. It is the ladder the empirical rule has to reproduce.

q	M	worst evolved %	worst all-times %	median GPU ms	converged
0	64	1.8890	2.5629	333.2	yes
16	128	1.3985	2.4806	417.9	yes
32	192	1.2336	2.3534	496.9	yes
64	320	1.0843	2.1489	701.7	yes
128	576	0.8930	1.8116	1330.4	yes
256	1088	0.5194	0.9053	4354.9	yes

Monotone on the evolved metric: ****yes****.

1.2.5 The rebuilt ladder

****EQ**, cheapest certified rule per rung****** – monotone on evolved times: ****no****; monotone on all times: yes; every rung converged: yes; cost within 2x of the old rule: yes; all-times not raised: yes; ****passes**: **no****.

q	M	m	$\rho_{\max}$	certified	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms
0	64	1024	0.0153	yes	1.8891	2.5629	2.5629	57.4
16	128	1024	0.0935	yes	1.4270	2.4806	2.4806	79.2
32	192	1024	0.0533	yes	1.2493	2.3534	2.3534	96.0
64	320	1024	0.0531	yes	1.2275	2.1489	2.1489	118.6
128	576	2048	0.1908	no	0.8925	1.8116	1.8116	246.8
256	1088	2048	0.1678	no	1.0361	1.0361	0.9053	699.3

****EQ**, incumbent static-population rule****** – monotone on evolved times: ****no****; monotone on all times: no; every rung converged: yes.

q	M	m	$\rho_{\max}$	certified	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms
0	64	256	0.2407	no	3.4166	3.4166	2.5629	50.8
16	128	512	0.1279	no	2.1427	2.4806	2.4806	71.6
32	192	768	0.1482	no	1.5026	2.3534	2.3534	90.4
64	320	1280	0.1439	no	1.1454	2.1489	2.1489	123.9
128	576	2048	0.2194	no	0.8928	1.8116	1.8116	245.4
256	1088	2048	0.4621	no	6.0676	6.0676	0.9053	688.6

****dense** (exact) quadrature****** – monotone on evolved times: ****yes****; monotone on all times: yes; every rung converged: yes.

1.3 R1 (control) – the field-metric ridge

1.3.1 Dense (exact) quadrature

**** $\lambda_{rel} = 0$ **** – monotone in q on evolved times: ****yes****; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; ****passes**: **yes****.

**** $\lambda_{rel} = 0.0001$ **** – monotone in q on evolved times: ****yes****; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; ****passes**: **yes****.

q	M	m	$\rho_{\max}$	certified	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	conv
0	64	–	–	–	1.8890	2.5629	2.5629	281.9	yes
64	320	–	–	–	1.0843	2.1489	2.1489	620.4	yes
256	1088	–	–	–	0.5194	0.9053	0.9053	3945.3	yes

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	cc
0	64	–	1.8890	2.5629	2.5629	295.2	1.000x	ye
16	128	–	1.3985	2.4806	2.4806	368.0	1.000x	ye
64	320	–	1.0843	2.1489	2.1489	627.2	1.000x	ye
256	1088	–	0.5194	0.9053	0.9053	3911.9	1.000x	ye

$\lambda_{rel} = 0.001$ – monotone in q on evolved times: **yes**; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; **passes**: **yes**.

$\lambda_{rel} = 0.01$ – monotone in q on evolved times: **yes**; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; **passes**: **yes**.

$\lambda_{rel} = 0.1$ – monotone in q on evolved times: **yes**; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; **passes**: **yes**.

$\lambda_{rel} = 1$ – monotone in q on evolved times: **yes**; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: no; **passes**: **no**.

1.3.2 Empirical quadrature, $m = 4M$

$\lambda_{rel} = 0$ – monotone in q on evolved times: **no**; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; **passes**: **no**.

$\lambda_{rel} = 0.0001$ – monotone in q on evolved times: **no**; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; **passes**: **no**.

$\lambda_{rel} = 0.001$ – monotone in q on evolved times: **no**; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; **passes**: **no**.

$\lambda_{rel} = 0.01$ – monotone in q on evolved times: **no**; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; **passes**: **no**.

$\lambda_{rel} = 0.1$ – monotone in q on evolved times: **no**; every rung converged: no; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; **passes**: **no**.

$\lambda_{rel} = 1$ – monotone in q on evolved times: **no**; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; **passes**: **no**.

1.4 R2 (control) – more tests at fixed q

1.4.1 Dense (exact) quadrature

$M = 16(K + q)$ – monotone in q on evolved times: **yes**; every rung converged: yes; cost within 1.5x of the incumbent: no; all-times metric not raised: yes; **passes**: **no**.

$M = 4(K + q)$ – monotone in q on evolved times: **yes**; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; **passes**: **yes**.

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	cc
0	64	–	1.8890	2.5629	2.5629	295.2	1.000x	ye
16	128	–	1.3983	2.4806	2.4806	368.7	1.002x	ye
64	320	–	1.0842	2.1489	2.1489	625.3	0.997x	ye
256	1088	–	0.5127	0.9053	0.9053	4020.5	1.028x	ye

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	cc
0	64	–	1.8890	2.5629	2.5629	295.2	1.000x	ye
16	128	–	1.3966	2.4806	2.4806	369.5	1.004x	ye
64	320	–	1.0830	2.1489	2.1489	627.4	1.000x	ye
256	1088	–	0.4727	0.9053	0.9053	3737.6	0.955x	ye

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	cc
0	64	–	1.8890	2.5629	2.5629	295.2	1.000x	ye
16	128	–	1.3838	2.4806	2.4806	369.9	1.005x	ye
64	320	–	1.0745	2.1489	2.1489	629.5	1.004x	ye
256	1088	–	0.4308	0.9053	0.9053	3032.5	0.775x	ye

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	cc
0	64	–	1.8890	2.5629	2.5629	295.2	1.000x	ye
16	128	–	1.3750	2.4806	2.4806	368.6	1.002x	ye
64	320	–	1.0877	2.1489	2.1489	633.7	1.010x	ye
256	1088	–	0.6251	0.9053	0.9053	2862.1	0.732x	ye

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	cc
0	64	–	1.8890	2.5629	2.5629	295.2	1.000x	ye
16	128	–	1.4297	2.4806	2.4806	366.9	0.997x	ye
64	320	–	1.2140	2.1489	2.1489	634.2	1.011x	ye
256	1088	–	1.1011	1.1011	0.9053	2657.6	0.679x	ye

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	
0	64	256	2.0659	2.5629	2.5629	51.8	1.000x	
16	128	512	2.1811	2.4806	2.4806	72.4	1.000x	
64	320	1280	1.1408	2.1489	2.1489	127.4	1.000x	
256	1088	2048	3.5204	3.5204	0.9053	715.4	1.000x	

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	
0	64	256	2.0659	2.5629	2.5629	51.8	1.000x	
16	128	512	2.1809	2.4806	2.4806	73.4	1.015x	
64	320	1280	1.1406	2.1489	2.1489	127.3	1.000x	
256	1088	2048	3.5155	3.5155	0.9053	732.1	1.023x	

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	
0	64	256	2.0659	2.5629	2.5629	51.8	1.000x	
16	128	512	2.1786	2.4806	2.4806	74.1	1.024x	
64	320	1280	1.1396	2.1489	2.1489	126.6	0.994x	
256	1088	2048	3.4804	3.4804	0.9053	669.1	0.935x	

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	
0	64	256	2.0659	2.5629	2.5629	51.8	1.000x	
16	128	512	2.1604	2.4806	2.4806	73.8	1.020x	
64	320	1280	1.1332	2.1489	2.1489	127.5	1.001x	
256	1088	2048	3.2834	3.2834	0.9053	518.4	0.725x	

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	
0	64	256	2.0659	2.5629	2.5629	51.8	1.000x	
16	128	512	2.1247	2.4806	2.4806	72.8	1.006x	
64	320	1280	1.1623	2.1489	2.1489	132.0	1.036x	
256	1088	2048	2.7294	2.7294	0.9053	485.4	0.679x	

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	
0	64	256	2.0659	2.5629	2.5629	51.8	1.000x	
16	128	512	2.1720	2.4806	2.4806	73.6	1.017x	
64	320	1280	1.3026	2.1489	2.1489	130.4	1.024x	
256	1088	2048	2.5988	2.5988	0.9053	452.3	0.632x	

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	conv
0	256	–	1.2710	2.5629	2.5629	343.1	1.209x	yes
16	512	–	1.2204	2.4806	2.4806	545.2	1.524x	yes
64	1280	–	1.0588	2.1489	2.1489	1301.8	2.095x	yes

**** $M = 8(K + q)$ **** – monotone in q on evolved times: ****yes****; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; ****passes: yes****.

1.4.2 Empirical quadrature, $m = 4M$

**** $M = 16(K + q)$ **** – monotone in q on evolved times: ****no****; every rung converged: yes; cost within 1.5x of the incumbent: no; all-times metric not raised: no; ****passes: no****.

**** $M = 4(K + q)$ **** – monotone in q on evolved times: ****no****; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; ****passes: no****.

**** $M = 8(K + q)$ **** – monotone in q on evolved times: ****no****; every rung converged: yes; cost within 1.5x of the incumbent: yes; all-times metric not raised: yes; ****passes: no****.

1.5 R3 – the weak residual on held-out test modes

Exactly integrated, in NumPy, from the retained per-step bank coefficients. The common held-out block is the 512 sine modes ranked 1537–2048, beyond every arm’s M in any of the three jobs. q-diag measured this on the dense fixed- $M = 256$ arms and found the on-test residual falling about 2x with q while the held-out residual rose about 1.7x, with the field error falling anyway; what is new here is the same measurement under a ****ridge**** and under ****larger M ****.

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	conv
0	64	–	1.8890	2.5629	2.5629	283.8	1.000x	yes
16	128	–	1.3985	2.4806	2.4806	357.7	1.000x	yes
64	320	–	1.0843	2.1489	2.1489	621.3	1.000x	yes

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	conv
0	128	–	1.4695	2.5629	2.5629	282.8	0.996x	yes
16	256	–	1.2305	2.4806	2.4806	419.3	1.172x	yes
64	640	–	1.0619	2.1489	2.1489	895.6	1.442x	yes

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	c
0	256	1024	1.3186	2.5629	2.5629	58.1	1.165x	y
16	512	2048	1.2202	2.4806	2.4806	108.5	1.526x	y
64	1280	2048	5.2481	5.2481	2.1489	184.3	1.485x	y

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	co
0	64	256	2.0659	2.5629	2.5629	49.9	1.000x	ye
16	128	512	2.1811	2.4806	2.4806	71.1	1.000x	ye
64	320	1280	1.1408	2.1489	2.1489	124.1	1.000x	ye

q	M	m	worst evolved %	worst all-times %	$t = 0$ compression %	median GPU ms	cost vs incumbent	co
0	128	512	1.5756	2.5629	2.5629	48.8	0.978x	ye
16	256	1024	1.8066	2.4806	2.4806	80.2	1.128x	ye
64	640	2048	1.1243	2.1489	2.1489	156.7	1.263x	ye

arm	q	quadrature	λ_{rel}	M	m	in-space	held-out (common block)	held/in
q0_m4_dense_l0	0	dense	0	64	–	1.57e-01	1.30e-01	0.829
q0_m4_eq_l0	0	eq	0	64	256	2.92e-01	1.33e-01	0.455
q0_m4_eq_l0_ret	0	eq	0	64	256	1.96e-01	1.34e-01	0.681
q16_m4_dense_l0	16	dense	0	128	–	1.27e-01	1.26e-01	0.994
q16_m4_dense_l0p0001	16	dense	1e-04	128	–	1.27e-01	1.26e-01	0.994
q16_m4_dense_l0p001	16	dense	1e-03	128	–	1.27e-01	1.26e-01	0.994
q16_m4_dense_l0p01	16	dense	1e-02	128	–	1.27e-01	1.26e-01	0.994
q16_m4_dense_l0p1	16	dense	1e-01	128	–	1.27e-01	1.27e-01	0.995
q16_m4_dense_l1	16	dense	1e+00	128	–	1.44e-01	1.27e-01	0.880
q16_m4_eq_l0	16	eq	0	128	512	1.35e-01	1.41e-01	1.043
q16_m4_eq_l0p0001	16	eq	1e-04	128	512	1.35e-01	1.41e-01	1.043
q16_m4_eq_l0p001	16	eq	1e-03	128	512	1.35e-01	1.41e-01	1.043
q16_m4_eq_l0p01	16	eq	1e-02	128	512	1.35e-01	1.41e-01	1.043
q16_m4_eq_l0p1	16	eq	1e-01	128	512	1.36e-01	1.41e-01	1.039
q16_m4_eq_l1	16	eq	1e+00	128	512	1.55e-01	1.41e-01	0.908
q256_m2_dense_l0	256	dense	0	544	–	2.85e-02	6.79e-02	2.383
q256_m4_dense_l0	256	dense	0	1088	–	3.13e-02	3.63e-02	1.160
q256_m4_eq_l0	256	eq	0	1088	2048	6.04e-02	5.76e-02	0.954
q64_m4_dense_l0	64	dense	0	320	–	9.56e-02	1.08e-01	1.127
q64_m4_eq_l0	64	eq	0	320	1280	9.56e-02	1.20e-01	1.255

arm	q	quadrature	λ_{rel}	M	m	in-space	held-out (common block)	held/in
q0_m16_dense_l0	0	dense	0	256	–	1.23e-01	1.15e-01	0.936
q0_m16_eq_l0	0	eq	0	256	1024	1.23e-01	1.22e-01	0.987
q0_m4_dense_l0	0	dense	0	64	–	1.57e-01	1.30e-01	0.829
q0_m4_eq_l0	0	eq	0	64	256	2.92e-01	1.33e-01	0.455
q0_m4_eq_l0_ret	0	eq	0	64	256	1.96e-01	1.34e-01	0.681
q0_m8_dense_l0	0	dense	0	128	–	1.37e-01	1.30e-01	0.945
q0_m8_eq_l0	0	eq	0	128	512	1.52e-01	1.36e-01	0.898
q16_m16_dense_l0	16	dense	0	512	–	1.01e-01	9.44e-02	0.933
q16_m16_eq_l0	16	eq	0	512	2048	1.07e-01	1.07e-01	1.003
q16_m4_dense_l0	16	dense	0	128	–	1.27e-01	1.26e-01	0.994
q16_m4_eq_l0	16	eq	0	128	512	1.35e-01	1.41e-01	1.043
q16_m8_dense_l0	16	dense	0	256	–	1.18e-01	1.15e-01	0.973
q16_m8_eq_l0	16	eq	0	256	1024	1.23e-01	1.32e-01	1.068
q64_m16_dense_l0	64	dense	0	1280	–	8.04e-02	6.84e-02	0.850
q64_m16_eq_l0	64	eq	0	1280	2048	9.60e-02	8.14e-02	0.848
q64_m4_dense_l0	64	dense	0	320	–	9.56e-02	1.08e-01	1.127
q64_m4_eq_l0	64	eq	0	320	1280	9.56e-02	1.20e-01	1.255
q64_m8_dense_l0	64	dense	0	640	–	8.63e-02	8.62e-02	1.000
q64_m8_eq_l0	64	eq	0	640	2048	9.43e-02	1.01e-01	1.070

arm	q	quadrature	λ_{rel}	M	m	in-space	held-out (common block)	held/in
q0_m4_dense	0	dense	0	64	–	1.57e-01	1.30e-01	0.829
q0_m4_eqcert	0	eq	0	64	1024	1.57e-01	1.28e-01	0.810
q0_m4_eqold_ret	0	eq	0	64	256	1.96e-01	1.34e-01	0.681
q0_m4_eqstatic	0	eq	0	64	256	1.85e-01	1.44e-01	0.778
q128_m2_eqold_bnd	128	eq	0	288	1152	8.09e-02	1.20e-01	1.488
q128_m4_eqcert	128	eq	0	576	2048	7.34e-02	9.76e-02	1.330
q128_m4_eqstatic	128	eq	0	576	2048	7.24e-02	9.61e-02	1.327
q16_m4_eqcert	16	eq	0	128	1024	1.27e-01	1.30e-01	1.023
q16_m4_eqstatic	16	eq	0	128	512	1.34e-01	1.38e-01	1.034
q256_m4_dense	256	dense	0	1088	–	3.13e-02	3.63e-02	1.160
q256_m4_eqcert	256	eq	0	1088	2048	5.54e-02	5.92e-02	1.070
q256_m4_eqstatic	256	eq	0	1088	2048	5.99e-02	4.40e-02	0.735
q32_m4_eqcert	32	eq	0	192	1024	1.19e-01	1.30e-01	1.088
q32_m4_eqstatic	32	eq	0	192	768	1.20e-01	1.35e-01	1.129
q64_m4_dense	64	dense	0	320	–	9.56e-02	1.08e-01	1.127
q64_m4_eqcert	64	eq	0	320	1024	9.91e-02	1.23e-01	1.237
q64_m4_eqstatic	64	eq	0	320	1280	9.56e-02	1.19e-01	1.249

control	worst all-times %	worst evolved %	median GPU ms	repetitions
nt1e-2_dt01	3.1999	3.1999	9.333	18
fft_loose	0.0338	0.0338	36.725	18
fft_tight	0.0000	0.0000	90.825	18

arm	q	M	m	worst all %	worst evolved %	median GPU ms	median iters	budget e
q0_m4_eq_l0_ret	0	64	256	2.5629	1.9002	50.4	3.0	0
q0_m4_eq_l0	0	64	256	2.5629	2.0659	51.8	3.0	0
q0_m4_dense_l0	0	64	–	2.5629	1.8890	295.2	3.0	0
q16_m4_dense_l0	16	128	–	2.4806	1.3985	368.0	3.0	0
q16_m4_eq_l0	16	128	512	2.4806	2.1811	72.4	3.0	0
q16_m4_eq_l0p0001	16	128	512	2.4806	2.1809	73.4	3.0	0
q16_m4_dense_l0p0001	16	128	–	2.4806	1.3983	368.7	3.0	0
q16_m4_eq_l0p001	16	128	512	2.4806	2.1786	74.1	3.0	0
q16_m4_dense_l0p001	16	128	–	2.4806	1.3966	369.5	3.0	0
q16_m4_eq_l0p01	16	128	512	2.4806	2.1604	73.8	3.0	0
q16_m4_dense_l0p01	16	128	–	2.4806	1.3838	369.9	3.0	0
q16_m4_eq_l0p1	16	128	512	2.4806	2.1247	72.8	3.0	0
q16_m4_dense_l0p1	16	128	–	2.4806	1.3750	368.6	3.0	0
q16_m4_eq_l1	16	128	512	2.4806	2.1720	73.6	3.0	0
q16_m4_dense_l1	16	128	–	2.4806	1.4297	366.9	3.0	0
q64_m4_eq_l0	64	320	1280	2.1489	1.1408	127.4	3.0	0
q64_m4_dense_l0	64	320	–	2.1489	1.0843	627.2	3.0	0
q64_m4_dense_l0p0001	64	320	–	2.1489	1.0842	625.3	3.0	0
q64_m4_eq_l0p0001	64	320	1280	2.1489	1.1406	127.3	3.0	0
q64_m4_dense_l0p001	64	320	–	2.1489	1.0830	627.4	3.0	0
q64_m4_eq_l0p001	64	320	1280	2.1489	1.1396	126.6	3.0	0
q64_m4_dense_l0p01	64	320	–	2.1489	1.0745	629.5	3.0	0
q64_m4_eq_l0p01	64	320	1280	2.1489	1.1332	127.5	3.0	0
q64_m4_eq_l0p1	64	320	1280	2.1489	1.1623	132.0	3.0	0
q64_m4_dense_l0p1	64	320	–	2.1489	1.0877	633.7	3.0	0
q64_m4_eq_l1	64	320	1280	2.1489	1.3026	130.4	3.0	0
q64_m4_dense_l1	64	320	–	2.1489	1.2140	634.2	3.0	0
q256_m2_dense_l0	256	544	–	0.9053	0.7566	3347.9	5.0	0
q256_m4_eq_l0	256	1088	2048	3.5204	3.5204	715.4	5.0	0
q256_m4_dense_l0	256	1088	–	0.9053	0.5194	3911.9	5.0	0
q256_m4_dense_l0p0001	256	1088	–	0.9053	0.5127	4020.5	4.5	0
q256_m4_eq_l0p0001	256	1088	2048	3.5155	3.5155	732.1	5.0	0
q256_m4_eq_l0p001	256	1088	2048	3.4804	3.4804	669.1	5.0	0
q256_m4_dense_l0p001	256	1088	–	0.9053	0.4727	3737.6	4.0	0
q256_m4_dense_l0p01	256	1088	–	0.9053	0.4308	3032.5	4.0	0
q256_m4_eq_l0p01	256	1088	2048	3.2834	3.2834	518.4	5.0	0
q256_m4_dense_l0p1	256	1088	–	0.9053	0.6251	2862.1	3.5	0
q256_m4_eq_l0p1	256	1088	2048	2.7294	2.7294	485.4	4.0	3
q256_m4_dense_l1	256	1088	–	1.1011	1.1011	2657.6	3.0	0
q256_m4_eq_l1	256	1088	2048	2.5988	2.5988	452.3	3.0	0

control	worst all-times %	worst evolved %	median GPU ms	repetitions
fft_loose	0.0338	0.0338	37.527	18
fft_tight	0.0000	0.0000	90.132	18
ntl-e-2_dt01	3.1999	3.1999	8.951	18

arm	q	M	m	worst all %	worst evolved %	median GPU ms	median iters	budget exits	v
q0_m4_dense_l0	0	64	–	2.5629	1.8890	283.8	3.0	0	9
q0_m4_eq_l0	0	64	256	2.5629	2.0659	49.9	3.0	0	9
q0_m4_eq_l0_ret	0	64	256	2.5629	1.9002	48.8	3.0	0	9
q0_m8_dense_l0	0	128	–	2.5629	1.4695	282.8	3.0	0	9
q0_m8_eq_l0	0	128	512	2.5629	1.5756	48.8	3.0	0	9
q0_m16_dense_l0	0	256	–	2.5629	1.2710	343.1	3.0	0	9
q0_m16_eq_l0	0	256	1024	2.5629	1.3186	58.1	3.0	0	9
q16_m4_eq_l0	16	128	512	2.4806	2.1811	71.1	3.0	0	9
q16_m4_dense_l0	16	128	–	2.4806	1.3985	357.7	3.0	0	9
q16_m8_dense_l0	16	256	–	2.4806	1.2305	419.3	3.0	0	9
q16_m8_eq_l0	16	256	1024	2.4806	1.8066	80.2	3.0	0	9
q16_m16_dense_l0	16	512	–	2.4806	1.2204	545.2	3.0	0	9
q16_m16_eq_l0	16	512	2048	2.4806	1.2202	108.5	3.0	0	9
q64_m4_eq_l0	64	320	1280	2.1489	1.1408	124.1	3.0	0	9
q64_m4_dense_l0	64	320	–	2.1489	1.0843	621.3	3.0	0	9
q64_m8_eq_l0	64	640	2048	2.1489	1.1243	156.7	3.0	0	9
q64_m8_dense_l0	64	640	–	2.1489	1.0619	895.6	3.0	0	9
q64_m16_dense_l0	64	1280	–	2.1489	1.0588	1301.8	3.0	0	9
q64_m16_eq_l0	64	1280	2048	5.2481	5.2481	184.3	3.0	0	9

1.5.1 R1 (qrg101)

1.5.2 R2 (qrg201)

1.5.3 EQCERT (qrg304)

1.6 Same-job full-order controls and every arm

1.6.1 R1 (qrg101) – full-order controls

1.6.2 R1 (qrg101) – every reduced-order arm

1.6.3 R2 (qrg201) – full-order controls

1.6.4 R2 (qrg201) – every reduced-order arm

1.6.5 EQCERT (qrg304) – full-order controls

control	worst all-times %	worst evolved %	median GPU ms	repetitions
fft_loose	0.0338	0.0338	36.684	18
fft_tight	0.0000	0.0000	89.689	18
ntl-e-2-dt01	3.1999	3.1999	8.800	18

arm	q	M	m	worst all %	worst evolved %	median GPU ms	median iters	budget exits
q0_m4_dense	0	64	–	2.5629	1.8890	281.9	3.0	0
q0_m4_eqcert	0	64	1024	2.5629	1.8891	57.4	3.0	0
q0_m4_eqold_ret	0	64	256	2.5629	1.9002	48.2	3.0	0
q0_m4_eqstatic	0	64	256	3.4166	3.4166	50.8	3.0	0
q16_m4_eqcert	16	128	1024	2.4806	1.4270	79.2	3.0	0
q16_m4_eqstatic	16	128	512	2.4806	2.1427	71.6	3.0	0
q32_m4_eqcert	32	192	1024	2.3534	1.2493	96.0	3.0	0
q32_m4_eqstatic	32	192	768	2.3534	1.5026	90.4	3.0	0
q64_m4_dense	64	320	–	2.1489	1.0843	620.4	3.0	0
q64_m4_eqcert	64	320	1024	2.1489	1.2275	118.6	3.0	0
q64_m4_eqstatic	64	320	1280	2.1489	1.1454	123.9	3.0	0
q128_m2_eqold_bnd	128	288	1152	1.8116	1.1515	184.0	3.0	0
q128_m4_eqcert	128	576	2048	1.8116	0.8925	246.8	3.0	0
q128_m4_eqstatic	128	576	2048	1.8116	0.8928	245.4	3.0	0
q256_m4_dense	256	1088	–	0.9053	0.5194	3945.3	5.0	0
q256_m4_eqcert	256	1088	2048	1.0361	1.0361	699.3	5.0	0
q256_m4_eqstatic	256	1088	2048	6.0676	6.0676	688.6	5.0	0

gate	passed	kind
artifacts_present	yes	blocking
backend_gpu	yes	blocking
bank_frozen	yes	blocking
bank_sha256_consistent	yes	blocking
checkpoint_unchanged	yes	blocking
complete	yes	blocking
cross_job_fidelity	yes	blocking
decoded_fields_match_saved_outputs	yes	blocking
directions_hash_matches_cclad01	yes	informational
directions_rank_covers_ladder	yes	blocking
eq_rules_untruncated	yes	blocking
evaluation_cohort_bitwise_abl01	yes	blocking
every_eq_rule_reports_validity	yes	blocking
every_invocation_paired	yes	blocking
every_rom_carries_exit_and_stationarity	yes	blocking
every_subject_case_has_all_reps	yes	blocking
final_cohort_unopened	yes	blocking
overdetermined_weak_system	yes	blocking
precision_highest	yes	blocking
r3_mode_blocks_complete	yes	blocking
recorded_errors_recomputed_from_saved_fields	yes	blocking
reference_fields_bitwise_match_a_source	yes	informational
reference_residuals	yes	blocking
repetition_output_identical	yes	blocking
reproduces_q0_m4_dense_l0	yes	blocking
reproduces_q0_m4_eq_l0_ret	yes	blocking
reproduces_q16_m4_dense_l0	yes	blocking
reproduces_q256_m2_dense_l0	yes	blocking
reproduces_q64_m4_dense_l0	yes	blocking
same_grid_baseline_present	yes	blocking
step_budget_600	yes	blocking
t0_field_invariant_in_lambda	yes	blocking
x64	yes	blocking

arm	source job	comparator arm	declared tolerance	relative difference (all-times)	relative d
q0_m4_dense_l0	btq101	q0_m4_dense_base	1e-09	9.53e-15	2.40e-13
q0_m4_eq_l0_ret	btq101	q0_m4_eq_base	1e-09	9.53e-15	4.05e-13
q16_m4_dense_l0	cclad01	q16_m4_dense_block	1e-09	2.40e-14	6.51e-15
q256_m2_dense_l0	btq102	q256_m2_dense_base_b600	1e-03	1.25e-13	3.64e-13
q64_m4_dense_l0	cclad01	q64_m4_dense_block	1e-09	9.51e-15	8.91e-14

gate	passed	kind
artifacts_present	yes	blocking
backend_gpu	yes	blocking
bank_frozen	yes	blocking
bank_sha256_consistent	yes	blocking
checkpoint_unchanged	yes	blocking
complete	yes	blocking
cross_job_fidelity	yes	blocking
decoded_fields_match_saved_outputs	yes	blocking
directions_hash_matches_cclad01	no	informational
directions_rank_covers_ladder	yes	blocking
eq_rules_untruncated	yes	blocking
evaluation_cohort_bitwise_abl01	yes	blocking
every_eq_rule_reports_validity	yes	blocking
every_invocation_paired	yes	blocking
every_rom_carries_exit_and_stationarity	yes	blocking
every_subject_case_has_all_reps	yes	blocking
final_cohort_unopened	yes	blocking
overdetermined_weak_system	yes	blocking
precision_highest	yes	blocking
r3_mode_blocks_complete	yes	blocking
recorded_errors_recomputed_from_saved_fields	yes	blocking
reference_fields_bitwise_match_a_source	yes	informational
reference_residuals	yes	blocking
repetition_output_identical	yes	blocking
reproduces_q0_m4_dense_l0	yes	blocking
reproduces_q0_m4_eq_l0_ret	yes	blocking
reproduces_q16_m4_dense_l0	yes	blocking
reproduces_q64_m4_dense_l0	yes	blocking
same_grid_baseline_present	yes	blocking
step_budget_600	yes	blocking
t0_field_invariant_in_lambda	—	blocking
x64	yes	blocking

1.6.6 EQCERT (qrg304) – every reduced-order arm

1.7 Gates

1.7.1 R1 (qrg101)

1.7.2 R1 – cross-job fidelity

1.7.3 R2 (qrg201)

1.7.4 R2 – cross-job fidelity

arm	source job	comparator arm	declared tolerance	relative difference (all-times)	relative difference
q0_m4_dense_l0	btq101	q0_m4_dense_base	1e-09	0.00e+00	2.43e-13
q0_m4_eq_l0_ret	btq101	q0_m4_eq_base	1e-09	0.00e+00	5.13e-14
q16_m4_dense_l0	cclad01	q16_m4_dense_block	1e-03	3.62e-14	2.13e-14
q64_m4_dense_l0	cclad01	q64_m4_dense_block	1e-03	3.31e-14	8.03e-14

1.7.5 EQCERT (qrg304)

gate	passed	kind
artifacts_present	yes	blocking
backend_gpu	yes	blocking
bank_frozen	yes	blocking
bank_sha256_consistent	yes	blocking
checkpoint_unchanged	yes	blocking
complete	yes	blocking
cross_job_fidelity	yes	blocking
decoded_fields_match_saved_outputs	yes	blocking
directions_hash_matches_cclad01	no	informational
evaluation_cohort_bitwise_abl01	yes	blocking
every_invocation_paired	yes	blocking
every_rom_carries_exit_and_stationarity	yes	blocking
every_rule_archived	yes	blocking
every_rule_reports_rho	yes	blocking
every_subject_case_has_all_reps	yes	blocking
final_cohort_unopened	yes	blocking
fit_and_certification_trajectories_disjoint	yes	blocking
no_rule_truncated	yes	informational
overdetermined_weak_system	yes	blocking
precision_highest	yes	blocking
r3_mode_blocks_complete	yes	blocking
recorded_errors_recomputed_from_saved_fields	yes	blocking
reference_residuals	yes	blocking
repetition_output_identical	yes	blocking
reproduces_q0_m4_eqold_ret	yes	blocking
reproduces_q128_m2_eqold_bnd	yes	blocking
rule_grid_complete	yes	blocking
same_grid_baseline_present	yes	blocking
step_budget_600	yes	blocking
x64	yes	blocking

1.7.6 EQCERT – cross-job fidelity

arm	source job	comparator arm	declared tolerance	relative difference (all-times)	relative difference
q0_m4.eqold.ret	btq101	q0_m4.eq_base	1e-09	0.00e+00	5.02e-13
q128_m2.eqold.bnd	btq101	q128_m2.eq_base	1e-03	2.47e-10	1.04e-09

1.8 Figures

One frozen Burgers checkpoint, 256 intervals, six development cases: the empirical-quadrature rule, certified on states the ROM reaches.

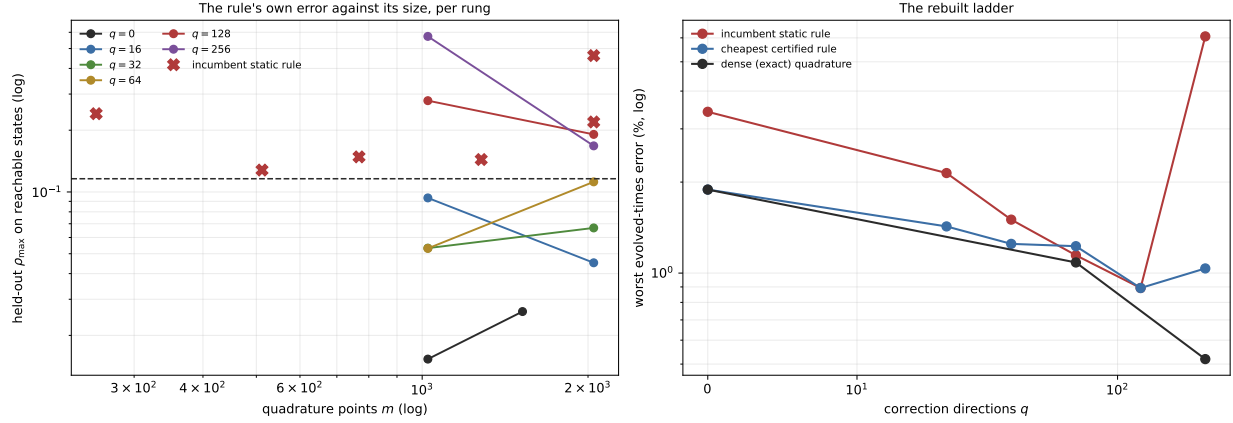


Figure 1: Left: the rule’s held-out ρ against its size m , per rung, with the declared bar. Right: the rebuilt ladder against the incumbent rule and the dense twins.

1.9 Glossary

- q — the number of extra correction directions added to the frozen neural head. The solved state is $u = G(h_\theta(z) + C_q y)$ with $z \in \mathbb{R}^{16}$ the latent code and $y \in \mathbb{R}^q$ the corrections. $q = 0$ is the incumbent reduced-order model.
- **rung** — one value of q in the ladder. "The ladder" is the sequence of rungs at a fixed rule for everything else.
- M — the number of weak test equations the per-step solve is graded on: M sine modes of the square, in ascending discrete-Laplacian order. The incumbent rule is $M = 4(K + q)$, four tests per unknown.
- m — the number of grid points the empirical-quadrature rule keeps when it approximates the advection integral. The dense arms use every interior point ($65 \setminus \{ \}, 025$ of them) and have no m .
- **dense / EQ quadrature** — dense evaluates the advection integral exactly on the grid; EQ replaces it with a nonnegative m -point rule fitted offline. Same unknowns, same tests, different integral.

The two demoted controls. Worst-over-evolved-times error against the same-job converged full-order solve.
Filled marker = every rung converged, open = not.

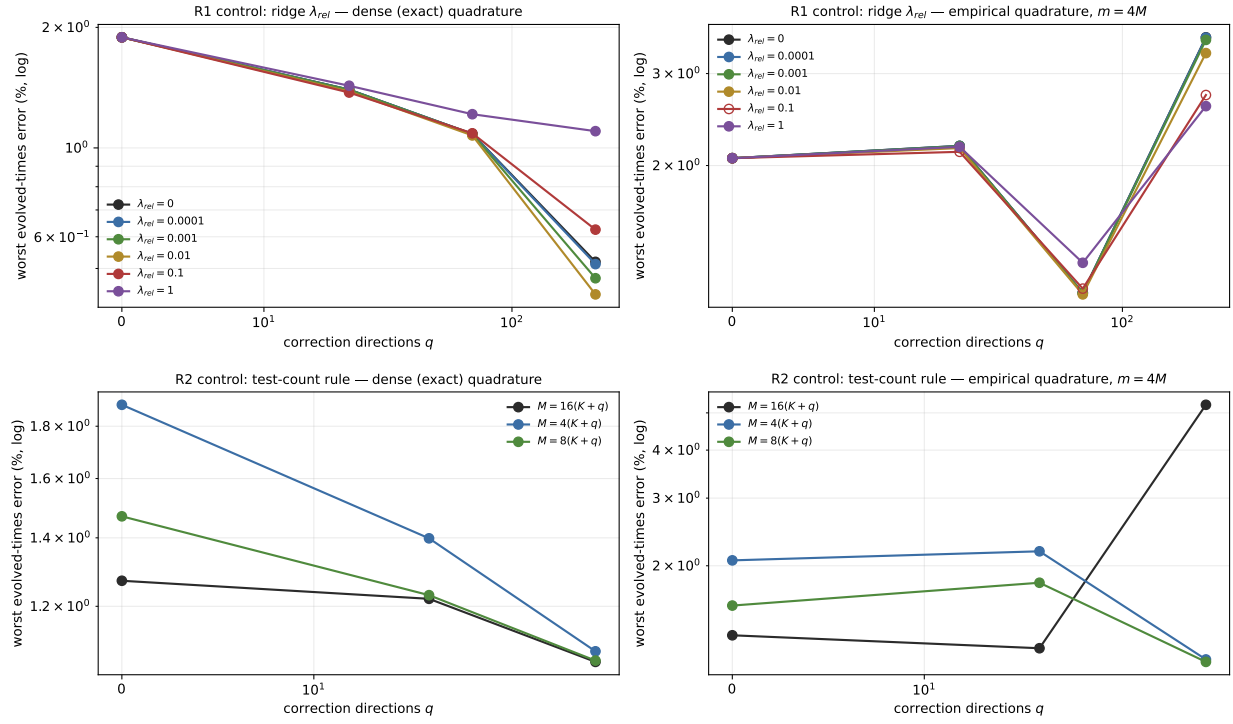


Figure 2: The two demoted controls: the ridge (top) and the test-count rule (bottom), dense (left) and empirical (right) quadrature.

- ρ — the rule’s own relative error on the advection functional it is the only approximation of: $\rho(u) = \|\sum_j w_j \Phi(x_j) a(u)(x_j) - \Phi^\top a(u)\| / \|\Phi^\top a(u)\|$. It is a property of a rule and a state, not of a solve.
- ρ^* , **the bar** — the value a rule’s worst held-out ρ must not exceed to be certified. Declared before the job ran at 0.116, the value the q-diag lane measured for the $q = 0$ incumbent rule at the state that carries the whole first-interval penalty.
- **reachable states** — states the reduced model actually visits: every Levenberg-Marquardt iterate of its own dense-quadrature rollouts on training-family trajectories. The incumbent rule is instead fitted on static decoder outputs, which is the difference this lane isolates.
- **held-out (rules)** — the certification states come from training-family trajectories disjoint from the ones the rule was fitted on, and both are disjoint from the six evaluation cases.
- **held-out test modes** — sine modes the solve never saw, used only afterwards to measure the weak residual. If a solve were overfitting its M tests, the residual on these would rise with q .
- **in-space residual** — the same weak residual on the arm’s own M modes, integrated exactly. For a converged dense arm it is essentially zero; for an EQ arm it is not, because the solve drove the quadrature-approximated residual to zero instead.
- **per-mode RMS** — a residual norm divided by the square root of the number of modes, so blocks of different size are comparable; normalised again by the per-node RMS of the previous state to make it dimensionless.
- λ_{rel} — the dimensionless ridge strength of the demoted R1 control. The solve minimises $\|r_w\|^2 + \lambda \|y\|_W^2$ with W the field metric and $\lambda = \lambda_{rel} \sigma_q^2$; $\lambda_{rel} = 0$ is the incumbent solve exactly.
- σ_q — the largest singular value of $\Phi^\top G C_q$, the exact linear response of the weak residual to the corrections. It is what makes λ_{rel} dimensionless and comparable across q , M and quadrature.
- **worst all-times error** — the largest relative error over the six output times, the six cases included; $t = 0$ is included, where the error is the decoder’s compression of the supplied initial field.
- **worst evolved-times error** — the same quantity over $t > 0$ only – the trajectory error proper, with the compression term removed.
- **$t = 0$ compression** — the relative error at $t = 0$: how well the reduced model can represent the initial field it was handed. It uses no quadrature at all, which is why every arm at a given q agrees there.
- **same-grid metric** — error measured against the same job’s own converged full-order solve (fft_tight) on the same 256-interval grid, so the discretisation error cancels.
- **converged** — every one of the 900 solved steps exited on the gradient, residual or tiny-step criterion – no step ran out of its 600-iteration budget – and the worst normalised joint gradient is at or inside 10^{-6} .

- **budget exit** — a time step that stopped because it hit the iteration budget rather than a convergence criterion. Any budget exit disqualifies a rung.
- **NNLS relative fit** — the residual of the nonnegative least-squares problem the rule is fitted by. It is reported but never used to certify a rule: it is flat across rules whose ρ differs by an order of magnitude.
- **truncated** — the fitter ran out of its declared walltime before reaching the target m . A truncated rule is disqualified.
- **fft_tight / fft_loose / nt1e-2_dt01** — same-job full-order Newton solves at three tolerance/time-step settings. `fft_tight` is the reference the same-grid metric is measured against, so its own error is zero by construction.
- **cross-job fidelity** — a named arm of this lane reproducing a named arm of an earlier audited job to a declared tolerance. It is how the lane proves it changed only what it said it changed.